

SAI SUBHASH KOTARU

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EDUCATION

New York University , Masters in Computer Science Relevant Courses - Artificial Intelligence, Deep Learning, Big Data, Database Management Systems	Exp Grad: May 2026
Indian Institute of Technology, Kharagpur , Dual degree in Electronics, Micro specialisation in AI Relevant Courses - ML, NLP, AI and Ethics, Statistics for AI/ML, Dependable and secure AI/ML	Aug 2018 – May 2023

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Bash, Matlab
Libraries & tools: PyTorch, Numpy, OpenCV, Hugging Face, TensorFlow, Docker, Git, FAISS, LangChain, LangGraph, vLLM, SGLang, LangSmith
Web/database/Cloud tools: FastAPI, React, SQL, Spark, MongoDB, Redis, Postgres, AWS, GCP, Lambda, Runpod

INDUSTRY EXPERIENCE

Allyin.ai (now Runara.ai) MLE Intern - LLM Inference optimisations, Agentic AI • Profiled prefill and decode bottlenecks using DCGM, identifying memory-bandwidth saturation in decode and SM underutilization in prefill • Designed RAG-based context pipeline from anomalies, telemetry, past incidents reducing root-cause analysis time from 20–40 min to 8 min • Built a multi-agent system for LLM inference triaging and optimization, where agents used tools over GPU telemetry, logs, and cloud APIs providing structured context to the diagnostic agent to diagnose issues and recommend fixes across heterogeneous GPU environments • Improved LLM serving efficiency through batch size tuning, concurrency, KV-cache allocation, reducing TCO per million tokens by 30%	June 2025 - Dec 2025
ZL Technologies Member of Technical Staff - Software Development • Developed 'Escalation' feature that improved Compliance module, reducing review cycle time by 35% via automated escalation • Worked on time-based partitioning and indexing of NoSQL database, which improved retrieval efficiency by 4% at petabyte scale • Reduced misflagging rate by 18% using finetuned BERT with class weighted loss and hard negative mining on near boundary samples	July 2023 - June 2024
Samsung RnD ML Research Intern - Recommendation Systems • Built a post-ranking optimization layer for IoT recommendations using Upper Confidence Bound to balance exploration and exploitation • Evaluated ranking quality using relevance and diversity, improved long-tail item exposure by 11%, keeping NDCG@50 within 5% of baseline	May 2022 - July 2022

ACADEMIC EXPERIENCE

ScalableLLM: AI Travel Planner (End-to-End ML Systems) 🔄 Agentic systems at scale, Inference optimisations, Context Management • Built a multi-agent LLM inference system - ReAct tool-use agent, planner agent, with LangSmith tracing and Prometheus monitoring, benchmarked under ramped concurrency (2-8), achieving 60% KV-cache hit rate, 197-674ms TTFT, and 10–15ms TPOT across 100 LLM calls • Architected inference layer optimizations including vLLM prefix caching, concurrent tool dispatch, API level caching, and context-window management (sliding-window trimming, brief compression), reducing prompt size 70% and eliminating context-overflow failures	Jan 2026 - Present
Masala CHAI (Data Synthesis, Agentic AI) Advanced Course Project under Prof. Ramesh Karri • Synthesized a 7.5k ImageToCode dataset, labelled using YOLOv11 for component detection achieving mAP 50 of 95.2%, mAP 50-95 83% • Implemented a ReAct style multi agent refinement system (component validation, netlist code generation, LLM-as-judge validation), with error recovery through iterative feedback loops and achieving 97.8% judge validation pass rate on human annotated test dataset • Engineered cost-optimized prompt caching and batching techniques achieving 15-18% LLM API cost reduction	Sep 2025 - Jan 2026
Layer selective rank reduction on VLMs 🔄 Deep Learning, Multimodal • Implemented layer-selective low-rank compression for CLIP, evaluating layer-wise tradeoffs between model efficiency and task quality • Achieved a 3% improvement in CIDEr scores on captioning COCO dataset, and 2% improvement on VQA dataset	Sept 2024 - Dec 2024
Fine Tuned RAG Systems for ROS2 Code generation (RAGs, MLOps) 🔄 Artificial Intelligence, Guide: Prof. Pantelis • Implemented a RAG pipeline with multistage retrieval query expansion, re-ranking, context selection to ensure grounded responses. • Supervised fine-tuned Llama-3.1-8B using LoRA + 8-bit quantization, enabling efficient on device inference and a fully deployed system	Oct 2024 - Dec 2024

PATENTS

Major Ravi Shanker Shukla, Sai Subhash Kotaru , Shailendra Kumar Varshney "A Multimode Fiber-based Intrusion detection system"	India Patent(Approved) 
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